Bank climate risk compliance and the construction carbon data gap

Banks face an urgent, regulation-driven need to report financed emissions for construction lending — yet no tool exists that converts project-level material data into PCAF-compliant output. CarbonIQ occupies a clear whitespace between construction LCA tools (which serve architects) and bank financed emissions platforms (which rely on sector-average proxies scoring PCAF 4–5). With **700+ PCAF signatory institutions**, carbonaccountingfinancials +2 the ECB already levying fines of up to **€7.6 million** for climate risk non-compliance, Compliance Corylated and ISSB S2 now mandated in **36 jurisdictions**, IFRS Foundation the regulatory pressure is intensifying globally. The **$16.5 trillion** global construction industry Research And Markets generates emissions that banks cannot accurately measure today — creating both a compliance crisis and a market opportunity worth pursuing across South and Southeast Asia, the GCC, and globally.

Executive summary

- **The regulatory wall is real and closing fast.** PCAF v3 (December 2025) Carbonaccountingfinancials now covers 10 asset classes ey Carbonaccountingfinancials including project finance, carbonaccountingfinancials ISSB S2 mandates financed emissions disclosure in 36 jurisdictions, IFRS Foundation and the ECB has begun issuing fines — €187,650 to ABANCA Disruption Banking Green Central Banking and €7.6 million to Crédit Agricole Compliance Corylated — for climate risk failures. HKMA, MAS, BNM, and CBSL all require or expect climate risk reporting from banks, with construction explicitly covered in multiple taxonomies.
- **Construction is the hardest asset class for banks to measure.** Most banks achieve PCAF data quality scores of **4–5 (lowest quality)** for construction lending, relying on sector-average proxies Manifest Climate that diverge from actual emissions by **100–200%**. Ox Unlike listed equities or mortgages where company or building-level data exists, construction project finance requires material-level embodied carbon data (A1–A5 lifecycle stages) that banks simply do not collect.
- **No existing platform bridges the gap.** Bank-focused tools (Persefoni, Watershed, MSCI) cannot process Bills of Quantities or calculate embodied carbon. Construction LCA tools (OneClick LCA, EC3, Tally) cannot output PCAF-compliant financed emissions. CarbonIQ would be the first platform to bridge these worlds — taking BOQ data as input, calculating A1–A5 embodied carbon/GWP, and outputting PCAF-compliant financed emissions with data quality scores of **2–3** instead of 4–5.
- **The addressable market is massive and growing.** The global construction market is **$16.5 trillion** (2025), Research And Markets the GCC alone has an **$819 billion** active construction pipeline, Blackridge Research and construction/real estate represents **10–24%** of bank loan books. The RegTech market is **$17–19 billion** growing at 16–23% CAGR, Precedence Research and PCAF Part C's December 2025 expansion to cover project insurance ey opens a parallel **$39 billion** construction insurance market.

- **DFCC Bank is a strong first-mover partner.** As issuer of Sri Lanka's first Green Bond (Daily FT) (LKR 2.5B), (NSE) first Blue Bond (LKR 3B), (Global Green Growth Institute) and first GSS+ Bond — with GCF accreditation granting access to **$250 million** in climate funding (DFCC Bank) — DFCC is ahead of the Sri Lankan market but does not yet report financed emissions under PCAF. Sri Lanka's mandatory SLFRS S2 adoption (beginning 2025 for top 100 listed entities) creates immediate demand. (Seneca ESG) Commercial Bank of Ceylon has signed an MOU with PCAF, signaling market readiness.

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1. Regulatory frameworks are converging on mandatory financed emissions disclosure

PCAF v3 sets the global measurement standard

The Partnership for Carbon Accounting Financials published its third edition on **2 December 2025**, (Esgbroadcast +2) now covering **10 asset classes** — adding use of proceeds structures, securitizations, and sub-sovereign debt to the original six. (carbonaccountingfinancials +2) The standard requires banks to report **absolute GHG emissions** (tCO2e) attributed to their lending and investment portfolios, (Persefoni) (Workiva) along with emissions intensity metrics and weighted average data quality scores per asset class. (GHG Protocol) (Carbonaccountingfinancials)

For **project finance** — the asset class most relevant to construction lending — PCAF requires banks to calculate financed emissions using the attribution formula: (Outstanding Amount / (Total Project Equity + Debt) × Project Emissions). (Open Risk Manual) (Workiva) The standard covers both operational and embodied carbon for construction, with embodied carbon (lifecycle stages **A1–A5**) required for new construction and major renovation loans. (Verco) (Department of Energy) PCAF acknowledges that its own building-type average datasets are "approximations due to variability in building composition, design, life cycle stages, and assumptions" and explicitly states that "actual data with higher quality scores (1 and 2) should be sought and used whenever possible." (Verco) (vercoglobal)

The data quality scoring system runs from **Score 1** (verified, independently audited project emissions) to **Score 5** (broad proxy estimates from EEIO tables). (Pulsora +2) For project finance specifically, Score 1 requires verified GHG data from the project itself, Score 3 requires physical activity-based data with credible emission factors, and Score 5 relies solely on outstanding loan amounts with sector-average factors. (Open Risk Manual) Over **700 financial institutions** across 70+ countries are now PCAF signatories, (carbonaccountingfinancials +6) with more than 250 having published disclosures. (Pulsora +3)

ISSB S2 mandates financed emissions for banks across 36 jurisdictions

IFRS S2 requires entities with commercial banking activities to disclose **absolute gross financed emissions, disaggregated by Scope 1, 2, and 3 for each industry by asset class**. (Grant Thornton) The December 2025 amendments provide some relief (IFRS Foundation) — banks may limit Scope 3 Category 15 to "financed emissions" only (excluding facilitated and insurance-associated emissions) (Grant Thornton +2) — but the core financed emissions requirement remains non-negotiable. (BDO) (PwC Viewpoint)

Adoption is accelerating: **Australia** made it mandatory (ISS-Corporate) from January 2025, (Nossadata) **Hong Kong** from August 2025 (comply-or-explain), (S&P Global) **Pakistan** from July 2025, (S&P Global) **Malaysia** mandatory and underway, **Qatar** from January 2026, (ISS-Corporate) and **Brazil** (Anthesis Global) and **Mexico** from 2026. (Nossadata) (ISS-Corporate) Sri Lanka adopted ISSB standards as SLFRS S1 and S2 with a phased timeline: top 100 CSE-listed entities from 2025, all listed entities by 2027, and companies above LKR 5 billion revenue by 2029.

Regional regulators are embedding climate risk into banking supervision

HKMA issued GS-1 climate risk management guidance in 2021 and launched a sustainable finance taxonomy in May 2024 (Hong Kong Government) that **explicitly covers construction** among its four Phase 1 sectors, (Moody's) referencing BEAM Plus green building certification criteria. (Hong Kong Monetary Authority) Phase 2A (September 2025) expanded to 25 activities across 6 sectors. (Hong Kong Monetary Authority)

MAS issued environmental risk management guidelines in December 2020 and new transition planning guidelines in **March 2026** (effective September 2027), requiring banks to assess climate transition pressures across sectors. Singapore's taxonomy covers real estate and construction.

Bank Negara Malaysia operates the Climate Change and Principle-based Taxonomy (CCPT) (Regulation Asia) with **mandatory half-yearly reporting since July 2022**, classifying exposures as Climate Supporting, Transitioning, or Watchlist. A dedicated VBIAF Sectoral Guide on **Construction & Infrastructure** (March 2022) addresses GHG emissions across the construction lifecycle. (IFAC)

CBSL published Sri Lanka's Green Finance Taxonomy in May 2022, (Greenfinanceplatform) explicitly covering the construction sector, (Greenfinanceplatform) and issued Direction No. 05 of 2022 requiring banks to report sustainable financing activities. The updated Roadmap for Sustainable Finance 2.0 launched in 2025. (Green Central Banking) (Sbfnetwork)

ECB has moved to active enforcement, issuing **binding supervisory decisions to 28 banks** in March 2023 (Green Central Banking) and now levying periodic penalty payments — **€187,650 against ABANCA** (November 2025) (Disruption Banking) and **€7.6 million against Crédit Agricole** (February 2026) (Compliance Corylated) — for failing to assess climate risk materiality. Penalties can accrue at up to **5% of daily average revenue** per day of non-compliance.

In the **GCC**, the UAE's landmark CB Law 2025 (Federal Decree-Law No. 6 of 2025) explicitly embeds ESG principles into the Central Bank's statutory objectives, with a one-year transition to September 2026. (GLA & Company) Saudi Arabia's regulatory framework remains largely voluntary, though SAMA is reportedly finalizing unified national ESG guidelines. (Ghazzawilawfirm)

ASEAN Taxonomy now covers construction

Version 3 of the ASEAN Taxonomy (effective December 2024) added **Technical Screening Criteria for Construction & Real Estate**, (Theacmf) covering construction and renovation of buildings, demolition, energy-efficient equipment, and building acquisition. (Ojk) The traffic-light classification (Green, Amber, Red) (Theacmf) benchmarks against international green building standards. (Ojk) (Rajah & Tann Asia) Version 4 (December 2024) expanded further to include agriculture, manufacturing, and professional services.

2. Banks face a structural data crisis in construction lending

The data simply does not exist in banks' systems

Financed emissions represent **at least 95% of banks' total carbon footprint** (Bain & Company +2) — yet for construction lending specifically, the data needed to measure these emissions accurately is fundamentally absent. Unlike listed equities (where companies report emissions), mortgages (where energy performance certificates exist), or motor vehicles (where fuel efficiency is known), construction projects produce unique, project-specific emissions profiles determined by material choices, quantities, and sourcing.

HSBC — one of the most advanced reporters globally — acknowledges that only **one-third of clients directly report Scope 1 and 2 emissions**, and only **10% of Scope 3 emissions** are available even in well-reported sectors. (Ox) For construction, reporting is almost certainly worse. HSBC's methodology note reveals they "do not differentiate between operation and construction" in project finance assessment, blending construction-phase embodied carbon with operational emissions. (HSBC)

Fifth Third Bank explicitly states its entire commercial and industrial portfolio sits at **PCAF data quality Score 5** — the lowest possible (Carbonaccountingfinancials) — and acknowledges these calculations "may not be useful for investment decisions." (Carbonaccountingfinancials) **BMO** (2025 disclosure) reports all borrowers at Score 5 and withholds absolute financed emissions figures because data quality is too poor to publish. (Bmo) **Wells Fargo** described its emissions data as "unverified, incomplete, or both." (Ox) **Citi** disclosed that in 2020, just **0.1% of Scope 3 data from power and energy clients was available**, with only 11% of any data independently verified. (Manifest Climate)

Sector-average proxies produce 100–200% errors

Banks currently rely on Environmentally Extended Input-Output (EEIO) models (GHG Protocol) — primarily from S&P Trucost — that assign a single emission factor per dollar of revenue for the "construction" sector. Research from the **Oxford Smith School** found that "nearly half of all methods of estimated emissions diverged from reported data by 100%, and over a quarter were off by at least 200%." (Ox) Triodos Bank and TD Bank both estimate **40–50% error margins** for Score 5 data. (Manifest Climate)

The fundamental failure of EEIO for construction is that it **cannot distinguish between a low-carbon mass-timber building and a high-carbon reinforced concrete tower** — both receive the same emission factor if their revenue is identical. A building designed for material efficiency using recycled steel and supplementary cementitious materials gets the same score as a building with excessive over-specification using virgin materials. This makes the data essentially useless for credit decisions, green loan structuring, or meaningful target-setting.

Why general AI cannot solve this

Banks cannot simply use ChatGPT or general-purpose AI for financed emissions calculations. Seven specific

barriers prevent this:

- **No auditability** — regulators require complete, traceable audit trails for every calculation (Manifest Climate)
- **Non-reproducibility** — the same query produces different outputs each time, violating regulatory consistency requirements (Manifest Climate)
- **Hallucination risk** — fabricated emission figures in regulatory submissions could constitute fraud (AgentiveAIQ)
- **No embedded PCAF methodology** — general AI does not follow PCAF's specific attribution formulas, scoring hierarchies, or asset-class calculation methods
- **No curated emission factor databases** — no access to material-specific embodied carbon factors (e.g., kgCO2e per tonne of C30 vs. C50 concrete)

No regulator — ECB, MAS, APRA, PRA, or HKMA — accepts AI-generated figures without transparent, documented methodology and external assurance capability.

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3. Construction lending represents a trillion-dollar market banks cannot accurately measure

Global construction is a $16.5 trillion industry

The global construction market reached **$16.45 trillion in 2025** and is projected to grow to **$21.7 trillion by 2030** at 5.9% CAGR. (Research And Markets) South Asia, Southeast Asia, and MENA were the fastest-growing regions in 2025, each exceeding 5% real growth. (Constructionowners) (Yahoo Finance) The global commercial lending market — where construction and real estate is the dominant sector — stood at **$10.92 trillion in 2025**, with Asia-Pacific commanding **48% market share**. (Fortune Business Insights)

Construction and real estate typically represents **10–24% of bank loan books**: large US banks average **~11%**, while regional and community banks average **~22%**. In Sri Lanka, construction accounted for **16.7% of private credit expansion** in April–October 2025, (Macrocolombo) and historically has reached 24% during lending booms. (Macrocolombo) Nearly **half of all US banks** report commercial real estate as their single largest loan category. (FAU Business) (Congress)

The GCC has an $819 billion active construction pipeline

Saudi Arabia alone has **5,200+ active construction projects** valued at **$819 billion**, representing 35% of all GCC project values. (Blackridge Research) Annual contract awards reached **$196 billion in 2025**, up 20% year-on-year. (Oxford Business Group) Vision 2030 has anchored over **$500 billion** in planned infrastructure investment. (GlobeNewswire) The GCC construction market overall is valued at **$175 billion (2025)** growing to **$214 billion by 2030**. (Mordor Intelligence)

Key mega-projects include **NEOM** ($500 billion total, (GlobeNewswire) Phase 1: $319 billion), (Oxford Business Group) **Diriyah Gate** ($63 billion), (Global Construction Review) **New Murabba** ($47 billion GDP

contribution), Blackridge Research and dozens of event-driven projects for the 2029 Asian Winter Games, 2034 FIFA World Cup, and Expo 2030.

Southeast Asian and Hong Kong construction remains robust

Singapore's EdgeProp.sg construction demand reached **SGD 50.5 billion (~$38 billion)** in 2025, Sgbc driven by Changi Terminal 5, MRT expansion, and Marina Bay Sands development. EdgeProp.sg BCA forecasts SGD 47–53 billion for 2026. Sgbc **Malaysia's** construction market is valued at **MYR 66 billion (~$15 billion)** in 2025, GlobeNewswire growing at 6–8%, with MIDA-approved investments of MYR 378.5 billion in 2024. GlobeNewswire **Hong Kong** saw HKD 417 billion ($54 billion) in property sales through October 2025.

Sri Lanka's construction sector is rebounding strongly

Construction grew **27.1% YoY in Q4 2024** and is expected to grow 7.9% in real terms in 2025, Business Wire driven by the Clean Sri Lanka Program (34 projects, LKR 17.8 billion), Business Wire a $3.7 billion Sinopec refinery project, U.S. Department of State and $200 million in ADB-funded power grid modernization. Business Wire The construction PMI hit **67.6 in September 2025** — its strongest reading since late 2021.

PCAF signatories are concentrated in target regions

Over **700 financial institutions** are PCAF signatories Carbonaccountingfinancials Carbonaccountingfinancials across 70+ countries, Carbonaccountingfinancials +4 including **DBS, OCBC, and UOB** in Singapore, Asiareengage CFP Green Buildings **HSBC** and international banks in Hong Kong, and a growing MENA region. Carbonaccountingfinancials In Sri Lanka, **First Capital Holdings** became the first capital markets institution to join PCAF, and **Commercial Bank of Ceylon** has signed an MOU to become a signatory. No Sri Lankan commercial bank is yet a full signatory — representing both a data point on market nascency and a first-mover opportunity.

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4. Where CarbonIQ plugs into the bank workflow

Step-by-step: from loan origination to PCAF annual report

The bank workflow for construction project finance follows ten distinct steps. CarbonIQ's primary insertion points are at loan origination/credit assessment (Steps 1–2) and annual financed emissions calculation (Steps 5–7):

Step 1 — Loan origination. The bank receives a construction project finance application. Basic data collected: borrower identity, project location, loan amount, project description, expected costs. **Carbon data is NOT currently collected at this step — this is the primary gap.** The bank has financial projections and project specs but no emissions data.

Step 2 — Credit assessment & due diligence. Traditional credit analysis plus environmental due diligence (where required). Environmental Impact Assessments may exist but rarely contain quantified embodied carbon.

➜ **CarbonIQ INSERTION POINT A:** The platform takes the project's Bill of Quantities (BOQ) — material types, volumes, specifications — which is always produced during project design, and calculates project-specific A1–A5 embodied carbon in tCO2e. This informs credit decisions, sustainability-linked loan structuring, and green bond eligibility assessment.

Step 3 — Loan approval & disbursement. CarbonIQ output becomes part of the loan documentation. For sustainability-linked loans, the embodied carbon baseline is embedded as a KPI.

Step 4 — Ongoing monitoring. Construction progress monitored. CarbonIQ can update calculations if material substitutions occur during construction.

Step 5 — Annual data collection. The ESG/sustainability team collects data across the entire lending portfolio. For each construction loan, they need borrower emissions data — which construction companies almost never provide. ➜ **CarbonIQ INSERTION POINT B:** The platform provides project-specific emissions data that the bank's systems currently cannot produce, filling the construction data void.

Step 6 — Financed emissions calculation. The bank applies the PCAF attribution formula: (Outstanding Amount / Total Project Equity + Debt) × Project Emissions. (Workiva) **CarbonIQ provides the "Project Emissions" variable** — currently estimated using sector-average proxies at Score 4–5. With CarbonIQ, this becomes verified BOQ-derived data at Score 2–3.

Step 7 — Data quality scoring. Each data point receives a PCAF score. (Stepchange) CarbonIQ-sourced construction data achieves **Score 2** (if independently verified) or **Score 3** (physical activity-based with standardized emission factors), versus the current **Score 4–5** baseline. This improvement is reported publicly in the bank's sustainability disclosures.

Steps 8–10 — Report production, external audit, and publication. CarbonIQ's transparent, reproducible methodology with full audit trail satisfies assurance requirements under ISAE 3000/3410. The bank publishes improved data quality scores, demonstrating regulatory compliance progress.

CarbonIQ output format and internal users

The platform outputs: **total project embodied carbon (tCO2e)**, broken down by material category and lifecycle stage (A1–A5), with the PCAF attribution factor pre-calculated based on the bank's loan share. This feeds directly into the bank's sustainability reporting system.

Bank team	How they use CarbonIQ
Credit/Lending	Assess project carbon risk at origination; structure green/sustainability-linked loans
Risk management	Climate stress testing; transition risk assessment; portfolio carbon concentration
ESG/Sustainability	Annual PCAF financed emissions calculation; target-setting; net-zero pathway tracking
Finance/Reporting	Regulatory disclosures (ISSB S2, CSRD, Pillar 3)
External auditors	Assurance-grade methodology with reproducible, traceable calculations

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5. No existing platform bridges construction data and bank reporting

Bank platforms cannot process material-level data

The competitive landscape reveals a clear, structural gap. **Persefoni** (valued ~$500–700M, $194M raised) Tracxn provides PCAF-compliant financed emissions Persefoni Software Finder but uses company-level reported data and EEIO sector averages — it cannot ingest BOQ data or calculate material-specific GWP. **Watershed** ($1.8B valuation, $170M raised, serving 4 of top 6 US banks) ESG Today launched AI-driven Product Footprints for supply chain analysis ESG News but not for construction project BOQs. GlobeNewswire +2 **MSCI's** Total Portfolio Footprinting covers 20,000+ issuers MSCI but operates at issuer level, not project level. SEC.gov **S&P Trucost** — PCAF's own founding data partner — uses EEIO models that assign the same emission factor to every dollar of construction revenue regardless of materials used. **Clarity AI** ($450M+ valuation) covers 70,000+ companies with AI-powered analytics but has no construction project capability. Crunchbase Dealroom.co

Construction tools cannot output PCAF data

On the other side, **OneClick LCA** (the world's leading building LCA platform), One Click LCA **EC3** (Building Transparency's free embodied carbon calculator with 17,000+ EPDs), Carbonleadershipforum **Tally** (Revit plug-in for building LCA), and **Sphera** (acquired by Blackstone for $1.4B, The Software Report potentially being sold for $3B) MarketScreener all calculate embodied carbon with high accuracy — but none integrates with bank reporting systems, applies PCAF attribution methodology, or outputs financed emissions in tCO2e per dollar invested.

The competitive gap in a single table

Capability	Bank platforms (Persefoni, Watershed, MSCI)	Construction tools (OneClick LCA, EC3)	CarbonIQ
BOQ input processing	❌	✅	✅
A1–A5 embodied carbon calculation	❌	✅	✅
PCAF attribution factor	✅	❌	✅
PCAF-compliant financed emissions output	✅	❌	✅
Bank-ready workflow integration	✅	❌	✅
Designed for financial institutions	✅	❌	✅

This gap exists because construction tech companies and financial tech companies operate in entirely separate ecosystems — different buyer personas, sales channels, technical communities, and regulatory knowledge. Converting a BOQ to A1–A5 GWP requires LCA expertise and material emission factor databases; converting that to PCAF-compliant financed emissions requires financial sector expertise and regulatory methodology. Few teams possess both capabilities.

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6. PCAF data quality scores reveal the upgrade opportunity

Banks currently score 4–5 for construction — the worst possible

Disclosed PCAF data quality scores from major banks confirm that construction and project finance consistently receive the lowest scores:

Bank	Asset class	PCAF Score	Year
Fifth Third Bank	Full C&I portfolio	**5**	2022
BMO	Full portfolio	**5**	2024
HSBC	Project finance (no project data)	**5**	2024
Vancity	Commercial real estate	**4.6**	2021
Aareal Bank	Commercial real estate (avg)	**3.2**	2024
Aareal Bank	CRE – US geography	**4.0**	2024

Climate Risk Review's analysis of 20 banks found that "most assigned data quality scores of 3–5 for the majority of their portfolios." Manifest Climate For construction lending specifically, banks universally report at Score 4–5 because construction companies rarely report emissions, Canadianlenders no standardized project-level data exists in bank systems, and the fallback is always sector-average EEIO factors.

Verified BOQ data would achieve Score 2–3 — a transformative upgrade

PCAF's data quality hierarchy for project finance is explicit:

- **Score 1:** Verified GHG emissions data calculated in accordance with the GHG Protocol Open Risk Manual Stepchange — achievable with independently audited, project-specific LCA from verified BOQ data with third-party-verified EPDs
- **Score 2:** Unverified but project-specific emissions data — achievable with CarbonIQ calculations from actual BOQ using standardized emission factor databases (ICE, PCAF materials dataset), without independent third-party audit
- **Score 3:** Physical activity-based estimates using known material quantities and credible emission factors
- **Score 4–5:** Current baseline using revenue proxies and EEIO tables Open Risk Manual

Moving from Score 5 to Score 2 reduces error margins from ~45–50% to ~10–15% Manifest Climate and crosses the threshold where data "has high informative value" Carbonaccountingfinancials for credit decisions and regulatory reporting. PCAF explicitly states that higher scores "should be sought and used whenever possible." Verco +2

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7. Insurance-associated emissions open a parallel market

PCAF Part C now covers construction project insurance

The December 2025 second edition of PCAF Part C expanded insurance-associated emissions methodology (Carbonaccountingfinancials) to cover **project insurance** (Construction All Risk and Erection All Risk policies) and **treaty reinsurance**, joining existing coverage of commercial insurance and personal motor insurance. (Carbonaccountingfinancials) (ey) Attribution follows the "follow the risk" principle (Carbonaccountingfinancials) based on premium as a proportion of project cost. (EY) The standard explicitly acknowledges "a lack of publicly available or free-to-use emissions data for construction projects." (EY)

The construction insurance market is $39 billion and growing at 9%

The global construction insurance market reached **$39.2 billion in 2024**, projected to grow to $92 billion by 2034 at 8.9% CAGR. (Market.us) The engineering insurance market (specifically CAR/EAR policies) is valued at **$26–32 billion**, growing at 9.5% CAGR. (Market Research Future) Asia-Pacific is the fastest-growing region. (Lucintel) (Allied Market Research) The GCC's $175 billion construction market generates an estimated **$3.5–5.3 billion** in annual construction insurance premiums alone.

Carbon risk is invisible in construction underwriting today

Construction insurers do **not systematically factor embodied carbon into underwriting or pricing** for CAR/EAR policies. As the Carbon Leadership Forum notes: "Embodied carbon is currently an invisible line item in real estate underwriting. It's there, but nobody sees it." (Carbonleadershipforum) BCIS confirms that "the reporting and measuring of whole life embodied carbon is currently not mandated for insurance companies" — but warns that carbon and material costs will become "equally as important." (BCIS)

AXA XL's SAMI (Sustainable Building Practices Adoption Maturity Index, launched March 2025) is the most advanced development — benchmarking contractors across carbon management, materials, and waste (ReinsuranceNe.ws) — but it evaluates contractor practices, not project-specific embodied carbon. No identified provider currently offers project-level embodied carbon data integrated into CAR/EAR underwriting workflows.

CarbonIQ data could serve insurers in four ways: PCAF Part C compliance (improving data quality from Score 5 to 2–3), underwriting risk scoring (carbon intensity as a supplementary risk factor), green premium design (discounts for low-carbon projects), and portfolio emissions reporting for annual ESG disclosures.

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8. DFCC Bank is positioned as the ideal Sri Lankan launch partner

Sri Lanka's green finance pioneer

DFCC Bank has issued Sri Lanka's **first Green Bond** (FINANCIAL CHRONICLE) (LKR 2.5 billion, September 2024, for renewable solar projects), (DFCC Bank) **first Blue Bond** (LKR 3 billion, November 2025, for marine

and water infrastructure), Global Green Growth Institute MarketScreener and **first Basel III-compliant GSS+ Bond** Businesstoday FINANCIAL CHRONICLE — all oversubscribed. DFCC's bonds are listed on the **Colombo Stock Exchange, Luxembourg Green Exchange, and NSE International Exchange at GIFT City**, DFCC Bank Daily FT giving the bank international green capital market presence unique among Sri Lankan institutions.

DFCC is the **only entity in Sri Lanka accredited by the Green Climate Fund**, providing access to **USD 250 million** in concessionary climate funding. MarketScreener DFCC Bank The bank established a comprehensive Sustainable Bond Framework (December 2025) covering green, blue, social, gender, and sustainability-linked bonds, with eligible projects required to comply with Sri Lanka's Green Finance Taxonomy. MarketScreener

DFCC does not yet report financed emissions — creating the opportunity

Despite its green finance leadership, **DFCC Bank is not a PCAF signatory and does not publicly report financed emissions**. The 2025 Annual Report includes SLFRS S1 and S2 disclosures with independent limited assurance, Businesstoday but uses transition provisions — meaning comprehensive financed emissions disclosure has not yet begun. MarketScreener The bank reports operational Scope 1 and 2 emissions and targets carbon neutrality by 2030. Cse

This creates a clear value proposition: CarbonIQ would enable DFCC to **become the first Sri Lankan bank to report PCAF-compliant financed emissions for its construction lending portfolio**, strengthening its green bond credibility, supporting GCF reporting requirements, and positioning it ahead of competitors as SLFRS S2 mandates expand.

The Sri Lankan context supports early adoption

Sri Lanka needs an estimated **$10.85 billion by 2030** to meet its Paris Agreement NDCs Green Central Banking Green Central Banking and approximately **$220 billion by 2053** for climate-resilient investments. World Bank The Central Bank's updated Roadmap for Sustainable Finance 2.0 (2025) is driving expanded taxonomy coverage and incentive frameworks. Green Central Banking Sbfnetwork SLFRS S2 becomes mandatory for all listed entities by 2027 Seneca ESG and all companies above LKR 5 billion revenue by 2029. S&P Global

Commercial Bank of Ceylon has signed an MOU with PCAF and is advancing financed emissions calculations with IFC support — indicating competitive pressure on DFCC to follow suit. **First Capital Holdings** became the first Sri Lankan capital markets firm to join PCAF. Construction accounts for approximately **17–24% of private credit expansion** in Sri Lanka, making it a material portfolio segment that cannot be excluded from financed emissions reporting.

The construction sector's strong rebound — **27.1% growth in Q4 2024**, Business Wire PMI of 67.6 in September 2025, and major projects including the $3.7 billion Sinopec refinery, U.S. Department of State Port City Colombo, and ADB-funded infrastructure — means DFCC's construction lending book is growing precisely as reporting requirements tighten.

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9. Key regulatory documents and reference links

Document	Organization	Link
PCAF v3 Part A — Financed Emissions (Dec 2025)	PCAF	https://carbonaccountingfinancials.com/files/standard-launch-2025/PCAF-PartA-2025-V3.pdf
PCAF Part C — Insurance-Associated Emissions (Dec 2025)	PCAF	https://carbonaccountingfinancials.com/files/standard-launch-2025/PCAF-PartC-2025-V2.pdf
PCAF Standard landing page	PCAF	https://carbonaccountingfinancials.com/standard
PCAF Signatories list	PCAF	https://carbonaccountingfinancials.com/en/signatories
PCAF Net-Zero Buildings Guidance	PCAF	https://carbonaccountingfinancials.com/files/downloads/pcaf-guidance-on-financing-the-net-zero-building-transition.pdf
PCAF Embodied Carbon Datasets	PCAF	https://carbonaccountingfinancials.com/financing-towards-net-zero-buildings
IFRS S2 Climate Disclosures Standard	ISSB	https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards-issb/english/2023/issued/part-a/issb-2023-a-ifrs-s2-climate-related-disclosures.pdf
IFRS S2 December 2025 Amendments	ISSB	https://www.ifrs.org/news-and-events/news/2025/12/issb-issues-targeted-amendments-ifrs-s2/
GHG Protocol	GHG Protocol	https://ghgprotocol.org/sites/default/files/2022-12/Chapter15.pdf

Document	Organization	Link
Scope 3 Category 15 Technical Guidance		
HKMA GS-1 Climate Risk Management	HKMA	https://www.hkma.gov.hk/eng/key-functions/banking/banking-regulatory-and-supervisory-regime/green-and-sustainable-banking/
HK Sustainable Finance Taxonomy (Phase 1)	HKMA	https://www.hkma.gov.hk/eng/news-and-media/press-releases/2024/05/20240503-3/
MAS ENRM Guidelines	MAS	https://www.mas.gov.sg/regulation/guidelines/guidelines-on-environmental-risk-management
MAS Transition Planning Guidelines (Mar 2026)	MAS	https://www.mas.gov.sg/news/media-releases/2026/guidelines-on-transition-planning
BNM Climate Change Principle-based Taxonomy	BNM	https://www.bnm.gov.my/documents/20124/938039/Climate+Change+and+Principle-based+Taxonomy.pdf
BNM VBIAF Construction & Infrastructure Guide	AIBIM/BNM	https://aibim.com/vbiaf-sectoral-guide/Vbiaf-Sectoral-Guide-on-Construction-and-Infrastructure.pdf
CBSL Sustainable Finance page	CBSL	https://www.cbsl.gov.lk/en/financial-system/financial-system-stability/sustainable-finance

Document	Organization	Link
Sri Lanka Green Finance Taxonomy	CBSL	https://www.cbsl.gov.lk/en/sl-green-finance-taxonomy
Sri Lanka National Climate Finance Strategy	Treasury/UNDP	https://www.treasury.gov.lk/api/file/ef2dff2b-8a19-4bc8-9936-10c3e17a9264
ASEAN Taxonomy v3/v4	ASEAN	https://asean.org/wp-content/uploads/2024/12/ASEAN-Taxonomy-Finalised-Version-3-4.pdf
UAE Climate Risk Principles	CBUAE	https://rulebook.centralbank.ae/en/rulebook/principles-effective-management-climate-related-financial-risks
DFCC Bank Green Bond	DFCC	https://www.dfcc.lk/products/dfcc-green-bond/
DFCC Annual Report 2024	DFCC	https://dfcc2024.annualreports.lk/

10. Market sizing and investment potential support a compelling venture case

The RegTech market is $17–19 billion and accelerating

The global RegTech market is estimated at **$17–19 billion (2025)**, growing at **16–23% CAGR** to reach $70–105 billion by 2030–2034. Asia-Pacific is the fastest-growing region at **~18.5% CAGR**, driven by fintech hubs in Singapore, Hong Kong, and Australia. Banking and financial services represent approximately **50% of end-user demand**. Banks paid a record **$19.3 billion in compliance penalties in 2024**, creating a strong value proposition for automation.

The ESG data market stands at **$5.87 billion**, growing at 24.9% CAGR to $17.8 billion by 2030. The pure-play carbon accounting software market is estimated at **$0.6–1.5 billion** in 2025, with Verdantix forecasting $1.5 billion by 2027. Growth drivers include California SB 253/261 (effective 2026–2027), EU CSRD (50,000+ companies), and ISSB S1/S2 adoption across 36 jurisdictions.

Comparable valuations show strong market appetite

Company	Valuation/Acquisition	Total Funding	Revenue (if available)
Watershed	**$1.8B** (Feb 2024 Series C)	~$170M	Not disclosed; 500+ customers including 4 of top 6 US banks
Persefoni	~$500–700M (estimated)	$194M	$4.3M (2024)
Clarity AI	**$450M+** (2021 Series B; likely higher post-2025 round)	$117–335M	Not disclosed; 70K+ companies covered
Sphera	**$1.4B** Blackstone acquisition (2021); potential **$3B** sale (2024–25)	N/A	N/A
Sustainalytics	**€170M** Morningstar acquisition (2020)	N/A	650+ employees at acquisition
MSCI ESG & Climate	Part of **~$48B** market cap entity	N/A	**$344M** annual run rate (ESG segment, 2024)
Greenomy	Early-stage	~€10–15M	EU-focused taxonomy platform

The **Sphera trajectory** is particularly instructive: acquired for $1.4B in 2021, now potentially valued at $3B — a **2.1x appreciation** in 3–4 years for an LCA/ESG data company.

Revenue model benchmarks for CarbonIQ

Based on comparable companies, three revenue models apply:

- **Enterprise SaaS subscription:** $100K–$500K annually for regional banks; $250K–$2M+ for global/Tier 1 banks. Multi-year contracts (2–3 years) preferred.
- **Per-project/usage-based:** Base platform fee plus per-project carbon assessment fees — particularly suited to CarbonIQ's model where each construction loan generates a discrete assessment.
- **Data licensing:** For aggregated, anonymized construction carbon data sold to insurers, rating agencies, and taxonomy providers — following the MSCI/S&P Trucost model with high margins and 94%+ retention.

Key investor criteria for climate RegTech platforms include: ARR growth (>100% YoY for early-stage), net revenue retention (>120%), blue-chip bank clients, regulatory moat (deep alignment with mandatory frameworks), audit-grade data quality, and clear path to profitability.

Competition pitch ammunition: the strongest facts for a FinTech/RegTech competition

The problem in one number: Banks' financed emissions from construction lending are measured with **100–200% error margins** using sector-average proxies that score PCAF data quality **4–5 out of 5** (worst possible). Oxford Smith School research confirms nearly half of all estimation methods diverge from reality by at least 100%.

The regulatory urgency: The ECB fined Crédit Agricole **€7.6 million** in February 2026 for climate risk failures. ISSB S2 is now mandatory in **36 jurisdictions** representing 60%+ of global GDP. PCAF has **700+ signatories**. There is no grace period left.

The market void: No existing platform combines BOQ input → A1–A5 embodied carbon → PCAF-compliant financed emissions. Bank tools (Persefoni, Watershed, MSCI) cannot process construction material data. Construction tools (OneClick LCA, EC3) cannot output bank-ready PCAF data. CarbonIQ sits at the only unoccupied intersection of a $16.5 trillion construction industry and a $17–19 billion RegTech market.

The data quality upgrade: CarbonIQ takes banks from PCAF Score 5 (sector proxies, ~45% error) to Score 2–3 (project-specific, ~10–15% error) — a **2–3 level improvement** that transforms construction lending from a reporting liability into a differentiator.

The dual market play: PCAF Part C (December 2025) expanded to cover project insurance (CAR/EAR policies), opening a parallel **$39 billion construction insurance market** that also lacks project-level carbon data. CarbonIQ serves both banks and insurers from one data layer.

The GCC pipeline alone justifies the platform: Saudi Arabia has **$819 billion** in active construction projects. Every one of these projects needs carbon data as PCAF signatories multiply and GCC regulators formalize ESG requirements (UAE CB Law 2025 embeds ESG in statute).

InsurTech angle: AXA XL launched its SAMI sustainability benchmarking tool for construction in March 2025 — but it benchmarks contractor practices, not project-specific carbon. No one is providing the actual embodied carbon numbers that insurers need for PCAF Part C compliance and risk-adjusted pricing.

Comparable exits validate the category: Sphera was acquired for $1.4B (2021) and is now reportedly valued at $3B. Watershed raised at $1.8B. The climate data market rewards platforms with regulatory moats and enterprise stickiness.

DFCC Bank proposal ammunition: Sri Lanka-specific findings

DFCC is a green finance leader without the data infrastructure to prove it. The bank has issued Sri Lanka's first Green Bond, first Blue Bond, and first GSS+ Bond, and is the country's only GCF-accredited entity with access to $250 million in climate funding. But DFCC does not yet report financed emissions under PCAF, meaning it cannot quantify the actual climate impact of its lending portfolio — undermining the credibility of its green bond programme with international investors.

A competitor is moving first. Commercial Bank of Ceylon has signed an MOU with PCAF to become an official signatory, supported by IFC. First Capital Holdings has already joined PCAF. If DFCC does not act, it risks losing its green finance leadership position in Sri Lanka.

SLFRS S2 makes this mandatory, not optional. Sri Lanka adopted ISSB standards with a phased timeline — top 100 listed entities from 2025, all listed entities by 2027. DFCC's 2025 Annual Report uses transition provisions; comprehensive financed emissions reporting cannot be deferred much longer. The bank needs methodology and tools to report financed emissions before the transition provisions expire.

Construction is ~17–24% of Sri Lankan private credit expansion. With the sector growing 27.1% YoY and major projects underway (Port City Colombo, Sinopec refinery, ADB-funded infrastructure), DFCC's construction lending book is expanding precisely as reporting requirements intensify. CarbonIQ would provide project-level carbon data for this material portfolio segment.

Green Climate Fund reporting demands granular data. As a GCF-accredited entity, DFCC must demonstrate measurable climate outcomes. Project-specific embodied carbon calculations — rather than sector-average estimates — would strengthen GCF reporting, justify continued concessionary funding access, and support green bond impact reporting for international exchanges (Luxembourg, GIFT City).

The CBSL taxonomy covers construction. Sri Lanka's Green Finance Taxonomy (May 2022) explicitly lists construction among covered sectors, and the updated Roadmap 2.0 (2025) is expanding taxonomy coverage. DFCC's Sustainable Bond Framework requires eligible projects to comply with the taxonomy. CarbonIQ would operationalize this compliance requirement by providing the construction-specific carbon data needed to verify taxonomy alignment.

The value proposition to DFCC: CarbonIQ enables the bank to (1) become the first Sri Lankan institution to report PCAF-compliant financed emissions for construction, (2) improve PCAF data quality from Score 5 to Score 2–3 for its construction portfolio, (3) strengthen green/blue/GSS+ bond impact reporting with project-level carbon data, (4) support GCF accreditation requirements with measurable climate outcome data, and (5) maintain green finance leadership ahead of Commercial Bank of Ceylon's PCAF entry.

Conclusion: CarbonIQ occupies a regulatory inevitability

The convergence of PCAF v3, ISSB S2, and regional regulatory mandates has created a world where banks **must** report construction financed emissions but **cannot** accurately measure them. This is not a hypothetical future state — the ECB is already levying million-euro fines, 36 jurisdictions have adopted ISSB S2, and 700+ institutions have committed to PCAF. The $16.5 trillion construction industry represents the largest, most opaque emissions category in bank lending portfolios.

CarbonIQ's positioning is not merely competitive but structurally necessary: it is the translation layer between the physical world of construction materials and the financial world of PCAF-compliant reporting. The December 2025 expansion of PCAF Part C to cover project insurance doubles the addressable market. The

GCC's $819 billion construction pipeline, Southeast Asia's fastest-growing RegTech demand, and Sri Lanka's emerging green finance market create a geographic expansion pathway that matches the competition's themes of RegTech automation and AI productivity.

The strongest insight from this research is that the gap CarbonIQ fills is not a feature gap in existing products — it is a **structural market failure** caused by the complete separation of construction technology and financial technology ecosystems. This means the barrier to competition is not purely technical but also requires dual-domain expertise that incumbents on neither side possess. That is a defensible moat.